[AI-02] AI Audit Report — HW02-AI Domain Testing

Course: Software Testing (FIT - VNUHCM University of Science)
Exercise ID: HW02-AI
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SUT: EShop (https://github.com/ttbhanh/eshop-sut)

> One batch generated by a single prompt = ONE artifact (one 5-section entry).

> Format inherited from HW01 `TEMPLATES/AI-02_Audit_Report.md`.

Artifact 0: HW02 init — report skeleton + reusable Domain-BVA skill (G9.2/G9.3 setup)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~10:30 16/10/2026 (UTC+07)
- **Full Prompt (user):**

```text
Tôi đã chuyển sang Build Mode. Hãy thực hiện Giai đoạn 1 (Khởi tạo dự án + Tạo Agent Skill + Dựng SUT):
1. Khởi tạo cấu trúc thư mục cho MSSV 23120372 (kế thừa HW01/TEMPLATES)...
2. Xây dựng Agent Skill skills/domain-bva-tester/ (4 bước)...
3. Ghi log & Commit số 1: chore(hw02): initialize report structure and create domain-bva-tester skill.
4. Khởi chạy SUT (kiểm tra Node, cài deps, node database.js, chạy backend 3000 / web 5173 / admin 5174).
Dừng lại sau khi hoàn tất để tôi kiểm tra trước khi chạy Domain Testing cho FR-02.
```

(2) AI Output

- Created `23120372_HW02_AI_DomainTesting/` with `Main_Report.md`, `Bug_Reports.md`, `AI_Audit_Report.md` (this file), `README.md`, `screenshots/.gitkeep`, `skills/domain-bva-tester/SKILL.md` + templates, `.gitignore`.
- No test cases generated yet — only skeletons with TODO markers for FR-02/09/16/20.
- SUT setup executed separately (see shell log in §Artifact 0 appendix / commit message).

(3) Verdict

VALID — skeleton matches HW02 §14 required contents (main report MD, bug report, audit+critique, commit log, README self-assessment) and reuses HW01 AI-02 5-section format. No hallucinated test results.

(4) Reasoning

Per ISTQB CTFL v4.0 §1.4 (test planning before design) and HW02 guiding principles (AI-First step-by-step + human review + git per step): initializing structure/skill first enforces stepwise workflow and auditability instead of one-shot generation.

(5) Student Fix

- Student to review file list, confirm folder name matches `<StudentID>_HW02_AI_DomainTesting_<grade>` convention at zip time, and confirm selected FRs (FR-02/09/16/20) do not duplicate group members before proceeding to FR-02 Domain step.

Artifact 1: FR-02 Domain Testing black-box (G9.2 Apply) — via domain-bva-tester Step 1

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~10:45 16/10/2026 (UTC+07)
- **Full Prompt (user):**

```text
Kích hoạt Agent Skill domain-bva-tester, thực hiện Step 1: Domain Testing (Black-box) cho FR-02:
chỉ đọc spec FR-02, tuyệt đối chưa đọc source code login; xác định domain variables
(email format/existence/case/spaces, password, failed_attempts, lockout_elapsed_time,
account_status); lập bảng Equivalence Partitioning Valid/Invalid + representative values
(dùng seed test@eshop.com); thiết kế TCs TC-FR02-DOM-01... vào Main_Report.md
(Actual/Status = Pending Execution); log Artifact 1; commit
"feat(FR-02): add black-box Domain Testing partitions and test cases".
```

(2) AI Output

- `Main_Report.md` §2.2: 8 domain variables (V1–V8), EP table (VP/IP + 2 spec-silent `SP` partitions for email case and whitespace), 15 TCs `TC-FR02-DOM-01…15` with partition mapping; boundary edges explicitly deferred to BVA (§2.3).
- `Main_Report.md` §6 + `README.md` §2: FR-02 Designed = 15 (rest 0).
- No source-code inspection performed; no execution results invented (all Status = Pending Execution).

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Partitions are disjoint, cover the FR-02 spec domain (format, existence, correctness, fail-counter state, lock state), and representatives use the real seed account. Two spec-silent partitions (case, whitespace) are correctly flagged rather than assumed.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (equivalence partitioning: divide input domain into partitions expected to be treated identically) and course Domain-Testing steps (variables → partitions → representatives → TCs, black-box before white-box): state variables (`failed_attempts` , `lockout_elapsed` , `account_status`) are treated as first-class domain inputs alongside field inputs, and exact edges are held back for BVA to avoid technique overlap.

(5) Student Fix

Student review decisions (16/10/2026), applied to `Main_Report.md` §2.2:

- **DOM-06 (empty password):** expectation fixed to validation error (HTTP 400 / form block) with NO counter increment; counting it toward lockout at runtime = defect.
- **DOM-11/DOM-12 ("consecutive"):** confirmed expectation that a 3rd-try success resets `failed_attempts` to 0; marked ★ Gap-analysis focus for code/execution cross-check.
- **SP3.1/SP4.1:** kept spec-silent classification, but student standard set to normalization (case-insensitive + auto-trim accept); actual behavior to be recorded at execution.

Artifact 2: FR-02 BVA black-box (G9.2 Apply) — via domain-bva-tester Step 2

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~11:00 16/10/2026 (UTC+07)

- **Full Prompt (user):**

```text
Kích hoạt domain-bva-tester, thực hiện Step 2 BVA cho FR-02 (vẫn black-box, chưa mở source):
boundaries + step unit cho failed_attempts (ngưỡng 3, step 1: mốc 0/1/2/3/4),
lockout_elapsed (ngưỡng 30s, step 1s: 0/1/15/29/30/31s, làm rõ <30 vs <=30),
độ dài input (rỗng/1 ký tự/7-8-9 ký tự/255-256+), biên kết hợp
(lần 3 đúng pass; hết 30s lại sai pass); bảng phân tích BVA 6/7-point + lý do từng điểm;
TCs TC-FR02-BVA-01... không trùng DOM; cập nhật Main_Report + Summary;
log Artifact 2 + chốt Artifact 1; commit "feat(FR-02): add Boundary Value Analysis (BVA) points and test cases".
```

(2) AI Output

- `Main_Report.md` §2.2: review decisions applied (DOM-06 validation-expectation, DOM-11/12 ★ focus, SP3.1/SP4.1 normalization standard).
- `Main_Report.md` §2.3: 5 boundaries with step units (B1 count T=3/u=1; B2 time T=30s/u=1s; B3 length u=1 char; B4 +1 exactness; B5/B6 multi-variable), 6/7-point analysis tables mapping each point to DOM or BVA TC, 13 BVA TCs `TC-FR02-BVA-01…13` (threshold crossing, no-extension at 4th, correct-at-3rd, 29/30/31s edges, nom 15s, wrong-at-expiry, 1-char, 7-char, 1000-char, double-submit).
- Summary: FR-02 Designed 15 → 28. No code inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Step-2 design holds: every ordered variable has an explicit step unit and each edge probes a concrete operator hypothesis. Two student decisions recorded: (a) ±1s timing tolerance accepted, with ms-precision server/DB timestamps required for operator conclusions; (b) BVA-11 (7-char) and BVA-12 (1000-char) probes kept to detect Register-rule leakage (`minlength=8`) and overflow/500 faults.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (boundary value analysis: test at and immediately either side of partition edges, including the 7-point min-/min/min+/nom/max-/max/max+ set) and the skill's step-unit rule: the 30s edge TC (BVA-06) is designed as an operator-determination probe rather than a pass/fail assertion, which is the correct black-box treatment when the spec leaves `<` vs `<=` undefined.

(5) Student Fix

- Student review decisions (16/10/2026): (a) ±1s tolerance accepted for BVA-05…09, execution must log ms-precision timestamps for operator conclusions; (b) BVA-11 and BVA-12 probes kept (Register-rule leakage + overflow check). No TC changes required; student approved opening source code for Step 3 (Gap Analysis).

Artifact 3: FR-02 white-box inspection + AI Gap Analysis (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~11:20 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3A — open EShop backend + frontend-web source; diff actual login/lockout logic (counter storage, comparison operators, expiry reset, 4th-while-locked, trim/lowercase) against the 28 black-box TCs; root-cause each miss (spec gap / LLM assumption / dev bug); add `TC-FR02-GAP-*` where needed; write `Main_Report.md` §2.4; finalize Artifact 2; log Artifact 3; commit `docs(FR-02): complete white-box inspection and AI gap analysis`.

(2) AI Output

- `Main_Report.md` §2.4: implementation table (11 rows with file:line refs), 8 misses M1–M8 with root causes, 9 supplementary TCs `TC-FR02-GAP-01…09`.
- Headline findings: `+2` increment (server.js:54), 180s lock (server.js:57), lock-on-2nd-fail, stale-counter relock, plaintext password in login response + storage, `type="text"` email/password inputs, swallowed 403 in UI. Two black-box hedges (SP3.1/SP4.1) confirmed correct.
- No execution performed yet; no Actual/Status filled.

(3) Verdict

VALID — reviewed and confirmed by execution, 16/10/2026. All 8 misses M1–M8 reproduced at runtime (counter +2, 180s lock, early lock, stale relock, password leak, no normalization, form types, feedback/i18n); code refs exact; GAP TCs map 1:1 to misses.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.3 (white-box/structural perspective complements black-box) and the course Gap-Analysis rubric (miss → why → human-added coverage): the analysis separates implementation facts (§2.4.1) from technique critique (§2.4.2) so the LLM's blind spots (response-shape domain, GUI-spec cross-check) are visible as process lessons, not just bug lists.

(5) Student Fix

- Execution confirms Artifact 3 in full (16/10/2026): 10 bugs FR02-BUG-01…10 filed in `Bug_Reports.md` + `Main_Report.md` §2.5. Two extra runtime discoveries beyond white-box reading: (a) first script run produced garbage DB reads — root cause was the server's fire-and-forget `db.run`, fixed with 600ms settle (methodology lesson for the report); (b) lost-update race under concurrent submits (FR02-BUG-10).
- Student severity review (16/10, applied): FR02-BUG-05 High→Critical, FR02-BUG-07 Medium→High, FR02-BUG-08 Low→Medium; BUG-09/10 stay Medium. GitHub Issue links pending (student files on group repo).

Artifact 4: FR-02 execution + screenshots + bug reports (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~11:40 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3B — execute all FR-02 TCs on live SUT (localhost:3000/5173) via automated script + headless browser; isolate/reset test-account state between cases; fill Actual/Status + summaries; screenshot each bug to `screenshots/FR02_BUG_*`; write `Bug_Reports.md` + `Main_Report.md` §2.5 in GitHub-Issue format; log Artifact 4; commit `test(FR-02): execute test cases, capture screenshots, and document bug reports`.

(2) AI Output

- `scripts/exec_fr02.js` (25 API steps, 8 fresh `exec*` users, sqlite inspection, real 15/29/30/31s probes, simulated expiry disclosed) → `scripts/exec_fr02_results.json`; `scripts/fr02_evidence.html` → `screenshots/FR02_API_EVIDENCE.png`; real page capture `screenshots/FR02_UI_LOGIN_PAGE.png` (headless Chrome).
- `Main_Report.md` §2.2.3/§2.3.3/§2.4.3: all 37 TCs filled — **13 PASSED, 24 FAILED**; §2.5: 10 bugs; summaries (Main + README): Designed 37 / Executed 37 / Passed 13 / Failed 24 / Bugs 10.
- `Bug_Reports.md`: 10 issues FR02-BUG-01…10 (env, pre-conditions, repro, expected/actual, severity, evidence, TC links). GitHub Issue links left TODO (group repo unknown — student files with PNGs).

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Statuses derive from measured transcript values (quoted verbatim); every FAILED maps to a filed bug; ±1s tolerance honored with ms timestamps; simulation and screenshot-rendering methods disclosed. Severity set finalized in review: BUG-05 Critical, BUG-07 High, BUG-08 Medium.

(4) Reasoning

Per ISTQB CTFL v4.0 §5 (test execution: compare actual vs expected, log defects with reproduction detail) and HW02 bug-reporting rules (Markdown + GitHub Issues + screenshots): API evidence is traceable step-by-step to the committed transcript, UI evidence is an unaltered page capture, and the one rendered PNG is generated from genuine output with its method stated — no fabricated results.

(5) Student Fix

- Student to review at next gate: confirm severity assignments (esp. BUG-07/09/10), file the 10 issues on the group repo and paste links into §2.5 + `Bug_Reports.md`, then approve Pool B (FR-09 Coupon). *(Review done 16/10: severities updated — BUG-05 Critical, BUG-07 High, BUG-08 Medium; GitHub links pending batch URLs from student.)*

Artifact 5: FR-09 Domain Testing black-box (G9.2 Apply) — via domain-bva-tester Step 1

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~12:00 16/10/2026 (UTC+07)
- **Full Prompt (user):** severity tune (BUG-05→Critical, BUG-07→High, BUG-08→Medium, GitHub links later) + run skill Step 1 Domain black-box for FR-09: spec + 4 seed codes only, no coupon/checkout source; variables from C1–C5 + money math (code text/existence, active, expiry incl. null, used/max, subtotal/min, per-user C5, percent-vs-fixed, over-discount); EP with VP/IP/SP (fixed > subtotal → 0 or negative? shipping in min_order?); 17 TCs `TC-FR09-DOM-*` (percent + fixed happy paths, one-condition violations C1→C5, spec-silent case/space/reapply/over-discount); update summaries; log Artifact 5 + finalize Artifacts 3–4; commit `feat(FR-09): add black-box Domain Testing partitions and test cases`.

(2) AI Output

- Severity updates applied in `Bug_Reports.md` + `Main_Report.md` §2.5 (counts unchanged, still 10 bugs).
- `Main_Report.md` §3.2: 9 variables (V1–V9, spec C-numbering C1 exists+active / C2 expiry / C3 min_order / C4 JWT / C5 max_uses), EP table (VP/IP + 5 SP: lowercase, spaces, stacking, repeat, rounding), 17 TCs `TC-FR09-DOM-01…17` — 3 happy-path math TCs with concrete VND figures (SAVE10 −400 000 → 3 600 000; BIGBUY → 3 950 000; VIP100 → 3 900 000), 6 single-violation TCs (one per C-condition incl. separate C1-nonexistent vs C1-inactive), quota TCs incl. VIP100 2×success + 3rd-reject, 5 SP record-as-is TCs. Boundary money/date/count edges deferred to BVA (§3.3).
- Summaries: FR-09 Designed = 17; Total 54 / Executed 37 / Passed 13 / Failed 24 / Bugs 10.
- No coupon/checkout source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026, with three review decisions applied: (a) VND figures kept; BVA to use u = 1 ₫ precision; (b) admin/DB fixtures approved for DOM-05/07/17 (declare + restore); (c) DOM-15/16 stance upgraded from record-as-is to normative single-code-per-order (overwrite-or-error; no double discount — stacking = serious bug).

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (each invalid partition exercised at least once; valid partitions combined into happy paths) and the domain-testing single-fault principle: DOM-04…09 each violate exactly one of C1–C5 while holding the other four, so any rejection unambiguously maps to its condition; quota TCs (DOM-09…11) additionally cover the per-user counting dimension of C5 across successive orders.

(5) Student Fix

- Student review decisions (16/10/2026): VND figures confirmed; u = 1 ₫ for BVA money edges; fixtures approved (DOM-05/07/17 + BVA setup, restore after run); DOM-15/16 upgraded to normative single-code expectation. Applied in §3.2 + §3.3.
- Student authorized white-box inspection (16/10); Artifact 6 verdict confirmed VALID above.

Artifact 7: FR-09 white-box inspection + AI Gap Analysis (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~12:40 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3A — open coupon/checkout source (backend + frontend-web); diff check order C1→C5, operators (min/max/expiry, string-vs-timestamp), `used_count` increment point + transaction/race, checkout trust (coupon_code/discounted total bypassing Apply → revalidation?), shipping-before/after-percent, fixed > subtotal floor; new misses → `TC-FR09-GAP-*`; write §3.4; finalize Artifact 6; log Artifact 7; commit `docs(FR-09): complete white-box inspection and AI gap analysis`.

(2) AI Output

- `Main_Report.md` §3.4: implementation table (13 rows with file:line refs), 8 misses M1–M8 with root causes, 11 supplementary TCs `TC-FR09-GAP-01…11`.
- Headline findings: percent formula `total×(1−value)` → negative discount / 10× final (server.js:398-421); C3 strict `>` (server.js:379); no auth on apply + client `user_id` (server.js:363-391); usage as voluntary post-checkout call, never enforced (server.js:297-309,443-454); checkout trusts any client total, ignores coupon (FR-08); UI `editableTotal` + trim/upper normalization (Checkout.jsx:14,29,91-103); no over-discount floor; no server-side single-code rule. Cross-FR notes queued for FR-10/11/12 phases.
- No execution performed yet; no Actual/Status filled.

(3) Verdict

VALID — reviewed and confirmed by execution, 16/10/2026. All 8 misses M1–M8 reproduced at runtime (percent math, `>` operator, missing auth, conditional usage check, client-trust checkout, normalization layers, no floor, no single-code rule); quota-lifecycle TCs (Q1–Q3) confirm the usage gate itself works when `user_id` is supplied — bounding the bypass bugs precisely.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.3 (structural perspective complements black-box) and the Gap-Analysis rubric: separating implementation facts (§3.4.1) from technique critique (§3.4.2) exposes the LLM's characteristic blind spots on this feature — formula-implementation errors, endpoint wiring/auth assumptions, and client↔server trust boundaries — as reusable process lessons.

(5) Student Fix

- Student to review at next gate: confirm GAP TC wording and cross-FR deferrals, then approve Chặng 3B (execution).
- Execution confirms Artifact 7 in full (16/10/2026): 10 bugs FR09-BUG-01…10 filed; runtime nuance (E2 skew) recorded; quota gate bounded (works iff `user_id` supplied).
- Execution confirms Artifact 7 in full (16/10/2026): 10 bugs FR09-BUG-01…10 filed in `Bug_Reports.md` + `Main_Report.md` §3.5. Runtime nuance beyond white-box reading: E2 hit T+17ms (HTTP skew) — exact-T operator stays code-determined (`expiry < now` = accept at T); quota gate verified working when `user_id` IS supplied (Q1–Q3/Q6), which bounds BUG-03/04/06 to their exact bypass shapes.

Artifact 8: FR-09 execution + screenshots + bug reports (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~13:00 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3B — execute all FR-09 TCs on live SUT via `scripts/exec_fr09.js` + UI checks (raw → `exec_fr09_results.json`; manage/reset VIP100 `used_count` for exact results); fill Actual/Status + summaries; per bug: UI

(`FR09_UI_*.png`) + API (`FR09_API_EVIDENCE.png`) screenshots; bugs `BUG-FR09-*` into `Bug_Reports.md` + §3.5; log Artifact 8; commit `test(FR-09): execute test cases, capture screenshots, and document bug reports` .

(2) AI Output

- `scripts/exec_fr09.js` (44 steps; fresh `cpn*` users; fixtures TIMEB1/FIX200/INACT1 created via admin endpoint — no role check needed — and DELETED after, seeds = 4 verified; real T−1/T/T+1 probes with skew logged; simulated nothing) → `scripts/exec_fr09_results.json` ; `scripts/fr09_evidence.html` → `screenshots/FR09_API_EVIDENCE.png` ; real capture `screenshots/FR09_UI_CHECKOUT.png` (editable total + coupon box).
- `Main_Report.md` §3.2.3/§3.3.3/§3.4.3: all 48 TCs filled — **28 PASSED, 20 FAILED**; §3.5: 10 bugs FR09-BUG-01…10; summaries (Main + README): Designed 85 / Executed 85 / Passed 41 / Failed 44 / Bugs 20.
- `Bug_Reports.md` : 10 issues (env, pre-conditions, repro, expected/actual, severity, evidence, TC links). GitHub Issue links left TODO (group repo unknown — student files with PNGs).

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Statuses derive from measured transcript values (quoted verbatim); every FAILED maps to a filed bug; ±1s/±1₫ tolerances honored (expiry skew logged per probe); fixture lifecycle evidenced in-transcript (F0/F9). Review decisions: BUG-07/09 → High (financial integrity / in-browser tampering surface); BUG-01/05 stay Critical; BUG-08/10 stay Medium.

(4) Reasoning

Per ISTQB CTFL v4.0 §5 (execution compares actual vs expected with reproducible defect records) and HW02 bug-reporting rules: quota-sensitive TCs use isolated users so `used_count` is exact (no cross-TC contamination); time/percent/floor edges assert computed money values rather than bare statuses, making operator/formula failures self-evidencing; the rendered PNG derives from genuine output with its method stated.

(5) Student Fix

- Student to review at next gate: confirm severity set (esp. BUG-07/08/09/10 Medium calls, BUG-01/05 Critical calls), file the 10 issues on the group repo and paste links, then approve Pool C (FR-16 CSV import). *(Review done 16/10: BUG-07/09 → High applied above; GitHub links pending; FR-16 Step 1 approved — import source stays closed.)*

Artifact 9: FR-16 Domain Testing black-box (G9.2 Apply) — via domain-bva-tester Step 1

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~13:20 16/10/2026 (UTC+07)
- **Full Prompt (user):** severity tune (BUG-07/09→High) + run skill Step 1 Domain black-box for FR-16: spec + admin-access rules only, no import source; 3-layer variables (file/request, CSV structure/RFC4180, row fields incl. name/price/stock/category, batch composition + atomic rollback); EP with VP/IP/SP (duplicate names? trailing `\n` ? spaces around commas?); 21 TCs `TC-FR16-DOM-*` with atomic-rollback focus; update summaries; finalize Artifacts 7–8; log Artifact 9; commit `feat(FR-16): add black-box Domain Testing partitions and test cases` .

(2) AI Output

- Severity updates applied in `Bug_Reports.md` + `Main_Report.md` §3.5 (counts unchanged, still 10 bugs).
- `Main_Report.md` §4.2: 11 variables across 3 layers (V1–V11, FR-12 roles + FR-15 field rules), EP table (VP/IP + 4 SP), 21 TCs `TC-FR16-DOM-01…21` — happy path with UTF-8 Vietnamese fixture, file/role/header/arity/quoting partitions, per-field invalid rows each asserting FULL rollback with cited reasons, ★ atomicity trio (first-bad / last-bad / commit-exactness with DB recounts), 3 SP record-as-is TCs (duplicates, trailing newline, spaced commas).

- **Header discrepancy handled:** SRS header has NO `stock` column — kept as SP10.1 (verify template at white-box/execution) instead of asserting it.
- Summaries: FR-16 Designed = 21; Total 106 / Executed 85 / Passed 41 / Failed 44 / Bugs 20.
- No import source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Atomicity is tested positionally (the only black-box way to separate validate-then-insert from insert-then-rollback); every invalid-row TC asserts both rollback AND error report; setup-free TCs use seeds only. Review decisions: atomicity trio confirmed as core; DOM-20/21 upgraded to normative best-practice (ignore trailing blank, trim spaces); duplicates stay observational vs FR-15; stock kept as SP probe; BVA dimensions approved.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (state-aware partitioning: batch composition is a domain input, not just row values) and the all-or-nothing requirement: DOM-17/18/19 form a minimal atomicity proof triple — first-bad detects fail-fast validation, last-bad detects missing rollback of committed prefixes, and the all-valid recount detects silent drops; any two alone leave one implementation shape untested.

(5) Student Fix

- Student to review at next gate: confirm atomicity triple, SP record-as-is stance (duplicates/trailing newline/spaces), stock-column verification plan, then approve BVA step (count/size edges).
- Review done 16/10 (decisions above, DOM-20/21 upgraded in §4.2); Artifact 9 verdict confirmed VALID.

Artifact 10: FR-16 BVA black-box (G9.2 Apply) — via domain-bva-tester Step 2

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~13:40 16/10/2026 (UTC+07)
- **Full Prompt (user):** BVA for FR-16 (still black-box): row-count edges (0-byte / header-only / 1 / 5 / 100 rows), error-index sweep in N = 5 (err@1/2/4/5, max = harshest), price edges u = 1 ₫ (−1 / 0-probe / 1 / 999999999), arity vs K = 5 (0 / K−1 / K / K+1 cols), name-length edges (1 / 255 / 256-1000 chars); BVA table + `TC-FR16-BVA-*` into §4.3; summaries; finalize Artifact 9; log Artifact 10; commit `feat(FR-16): add Boundary Value Analysis (BVA) points and test cases`.

(2) AI Output

- `Main_Report.md` §4.2: DOM-20/21 upgraded to normative best-practice (ignore trailing blank, trim spaces).
- `Main_Report.md` §4.3: 6 boundaries (B1 rows u=1 · B2 index u=1 · B3 price u=1₫ · B4 arity u=1 col · B5 length u=1 char · B6 stock probe), point-analysis tables, 20 BVA TCs `TC-FR16-BVA-01…20` — empty/header-only handling, C+N recounts, full err@1/2/4/5 sweep, price 0-probe (`> 0` vs `>= 0`), blank/K−1/K+1 arity, 255 vs 256/1000 length vs FR-15, stock-column probe. Fixture-dependent TCs declare setup; large batch notes timing.
- Summaries: FR-16 Designed 21 → 41; Total 126 / Executed 85 / Passed 41 / Failed 44 / Bugs 20.
- No import source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. The err@1/2/4/5 sweep covers every rollback-prefix shape; price-0/stock probes stay determination probes; length edges cross-check FR-15; scale TC watches timeout/truncation. Review: sweep, probes and scale TC confirmed; import source opening authorized; design TCs unchanged by review.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (boundaries on every ordered dimension, including structural ones like position-in-batch) and the skill's step-unit rule: positional atomicity cannot be covered by a single "one bad row" TC — the implementation could validate-then-insert (any position behaves identically) or insert-then-rollback (late errors expose half-commits), and only the full index sweep distinguishes them at black-box level.

(5) Student Fix

- Student to review at next gate: confirm index sweep, price-0/stock probes, 100-row scale TC, then approve opening import source for Step 3 (Gap Analysis).
- Review done 16/10 (decisions above); Artifact 10 verdict confirmed VALID.

Artifact 11: FR-16 white-box inspection + AI Gap Analysis (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~14:00 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3A — open import/product source (backend endpoint + upload middleware + CSV→SQLite logic, frontend-admin component); diff transaction/rollback, CSV parser (RFC4180 lib vs hand-split, CRLF/LF, trailing blanks), per-column validation (incl. `stock`, `price <= 0`/empty, `category_id` FK), JWT + `role === 'admin'` on import; new blind spots → `TC-FR16-GAP-*`; write §4.4; finalize Artifact 10; log Artifact 11; commit `docs(FR-16): complete white-box inspection and AI gap analysis`.

(2) AI Output

- `Main_Report.md` §4.4: implementation table (13 rows with file:line refs), 8 misses M1–M8 with root causes, 11 supplementary TCs `TC-FR16-GAP-01…11`.
- Headline findings: NO server file handling (JSON `{products}` contract, server.js:198-199; no csv/multer deps); NO transaction — partial-success commits (server.js:206-241); validation = only `!row.name` (price/category/stock unchecked, `|| 1` / `|| ""` defaults); no FK/UNIQUE/length caps, no `stock` column; no role check on any admin route (FR-12 systemic); UI naive `split(",")` parser with aliases, `price || 0`, `parseInt→1`, no extension check (App.jsx:364-418). Cross-FR notes queued (search-SQLi, even-id price string, cancel-from-shipping, unauth order read, canceled→delivered, missing roles).
- No execution performed yet; no Actual/Status filled.

(3) Verdict

VALID — reviewed and confirmed by execution, 16/10/2026. All 8 misses M1–M8 reproduced at runtime (no-file/no-transaction architecture, name-only validation, missing roles, tolerant garbage-in, layer misattribution resolved, stock-column dropped, partial-report semantics); commit-exactness TCs (D19/B03–B05) confirm the insert path itself is exact — bounding every bug to validation/atomicity/auth, never to counting.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.3 and the Gap-Analysis rubric: the analysis separates the architectural reframing (§4.4.1 — the endpoint is JSON-in, not CSV-in) from technique critique (§4.4.2), because most black-box TCs don't fail on values but on a wrong architectural assumption — the highest-value lesson of this feature for the AI Critique.

(5) Student Fix

- Student to review at next gate: confirm GAP TC wording and cross-FR deferrals, then approve Chặng 3B (execution).
- Execution confirms Artifact 11 in full (16/10/2026): 10 bugs FR16-BUG-01…10 filed; runtime extras (NULL-price acceptance, whitespace leftover id 17, 100-row 1684ms timing) recorded.
- Execution confirms Artifact 11 in full (16/10/2026): 10 bugs FR16-BUG-01…10 filed in `Bug_Reports.md` + `Main_Report.md` §4.5. Runtime extras beyond white-box reading: (a) NULL-price acceptance (missing key → NULL, not just unvalidated values); (b)

whitespace-name leftover id 17 (cleanup gap closed manually, script hardened with `TRIM(name) = ''`); (c) 100-row batch in 1684ms (no perf issue — scale TC passes).

Artifact 12: FR-16 execution + screenshots + bug reports (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~14:30 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3B — execute all FR-16 TCs on live SUT (localhost:3000/5174) via `scripts/exec_fr16.js` (CSV payloads per TC: 0-byte/header-only/blanks/arity/UTF-8/err@1…5/100 rows); DB recounts per TC for rollback/half-commit verdicts; clean test products afterwards; fill Actual/Status + summaries; UI (`FR16_UI_*.png`) + DB/API (`FR16_API_EVIDENCE.png`) screenshots; bugs into `Bug_Reports.md` + §4.5; log Artifact 12; commit `test(FR-16): execute test cases, capture screenshots, and document bug reports`.

(2) AI Output

- `scripts/exec_fr16.js` (45 steps; admin + customer + guest tokens; UI-parser replication per App.jsx:367-403 for file-layer TCs; per-TC DB recounts with 600–800ms settle; `T16*` cleanup + manual whitespace-leftover removal, count verified back to 5 seeds) → `scripts/exec_fr16_results.json`; `scripts/fr16_evidence.html` → `screenshots/FR16_API_EVIDENCE.png`; real capture `screenshots/FR16_UI_ADMIN.png` (admin login gate — fresh browser; import section evidenced via code refs, disclosed).
- `Main_Report.md` §4.2.3/§4.3.3/§4.4.3: all 52 TCs filled — **19 PASSED, 33 FAILED**; §4.5: 10 bugs FR16-BUG-01…10; summaries (Main + README): Designed 137 / Executed 137 / Passed 60 / Failed 77 / Bugs 30.
- `Bug_Reports.md`: 10 issues (env, pre-conditions, repro, expected/actual, severity, evidence, TC links). GitHub Issue links left TODO (group repo unknown — student files with PNGs).

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Statuses derive from measured transcript values and DB recounts (quoted verbatim); every FAILED maps to a filed bug; file-layer TCs proven via UI-parser replication (code-cited); cleanup verified (5 seeds), leftovers disclosed. Review decisions: BUG-02→Critical, BUG-08→Medium.

(4) Reasoning

Per ISTQB CTFL v4.0 §5 and HW02 bug-reporting rules: atomicity verdicts rest on DB counts, not response messages (the response correctly reports partial counts — the defect is that partial commits exist at all); the split between UI-flow evidence (replication + code refs) and API-flow evidence (live transcript) is labeled per TC so no UI claim rests on API data or vice versa.

(5) Student Fix

- Student to review at next gate: confirm severity set (esp. BUG-05/06/07 Medium calls, BUG-08/09/10 Low calls), file the 10 issues on the group repo and paste links, then approve Pool D (FR-20 Mobile). *(Review done 16/10: BUG-02→Critical, BUG-08→Medium applied; GitHub links pending; FR-20 Step 1 approved — mobile/backend source stays closed.)*

Artifact 13: FR-20 Domain Testing black-box (G9.2 Apply) — via domain-bva-tester Step 1

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~15:00 16/10/2026 (UTC+07)
- **Full Prompt (user):** severity tune (BUG-01/02 Critical, BUG-03/04 High, BUG-05/06/07 + BUG-08 Medium, BUG-09/10 Low) + run skill Step 1 Domain black-box for FR-20: FR-20 + FR-10 + FR-21–24 only, no mobile/backend source; state-machine + form variables (status, action, ownership, cancel-reason, network/feedback); EP (valid/invalid transitions, cross-user, SP: restore?

stock/coupon refund? double-tap); 18 TCs `TC-FR20-DOM-*` with transition matrix; update summaries; finalize Artifacts 11–12; log Artifact 13; commit `feat(FR-20): add black-box Domain Testing partitions and test cases`.

(2) AI Output

- Severity updates applied in `Bug_Reports.md` + `Main_Report.md` §4.5 (counts unchanged, still 10 bugs).
- `Main_Report.md` §5.2: 6 variables (V1–V6, FR-10/11/20–24), full 5×3 transition matrix, EP table (VP/IP + 4 SP), 18 TCs `TC-FR20-DOM-01…18` — pending/confirmed cancel-valid, shipping/delivered/canceled reject, cross-user ★, guest gate, dialog/dismiss/loading/empty-state/detail, 5 SP record-as-is (restore, coupon refund, stock restore, double-tap ★, reason field).
- **Transition correction:** brief grouped `confirmed` with invalid cancels — corrected per FR-10/FR-20 (`confirmed → canceled` VALID, own TC DOM-02); banner placed at §5.2 head.
- Summaries: FR-20 Designed = 18; Total 155 / Executed 137 / Passed 60 / Failed 77 / Bugs 30.
- No mobile/backend source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. The matrix leaves no (state × action) cell untested; cross-user/guest TCs isolate auth from state; SP TCs stay record-as-is where silent. Review decisions: `confirmed→cancel` correction confirmed (banner kept); SP upgraded to normative (stock +quantity AND coupon used−1 on cancel; debounce + strict server guard for double-tap); hybrid execution approved (API/concurrency via script + mobile UI via Expo-web/headless viewport, no physical phone).

(4) Reasoning

Per ISTQB CTFL v4.0 §4.3 (state-transition testing: valid transitions + invalid transitions + state coverage) and the domain single-fault principle: DOM-01…08 walk each matrix row exactly once with ownership/auth held constant, so a failure maps unambiguously to its transition; correcting the brief's `confirmed` grouping before designing (rather than after execution) prevents a full set of false failures.

(5) Student Fix

- Student to review at next gate: confirm transition matrix (esp. confirmed→cancel VALID), SP record-as-is stance, Expo-dependence note for UI TCs, then approve BVA step.
- Review done 16/10: matrix confirmed; DOM-15/16/17 upgraded to normative (refund + restore + debounce); hybrid execution plan approved. Artifact 13 verdict confirmed VALID.

Artifact 14: FR-20 BVA black-box (G9.2 Apply) — via domain-bva-tester Step 2

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~15:20 16/10/2026 (UTC+07)
- **Full Prompt (user):** BVA for FR-20 (still black-box): state edges (last-valid `confirmed`, first-invalid `shipping`, immediate re-cancel), time/race (ship-vs-cancel window, double-tap 0–300 ms), ownership edges (own vs `±1`/garbage ids), list edges (0/1/many); BVA table + `TC-FR20-BVA-*` into §5.3; summaries; finalize Artifact 13; log Artifact 14; commit `feat(FR-20): add Boundary Value Analysis (BVA) points and test cases`.

(2) AI Output

- `Main_Report.md` §5.2: DOM-15/16/17 upgraded to normative (refund + restore + debounce/guard).
- `Main_Report.md` §5.3: 7 boundaries (B1 validity edge · B2 re-cancel timing · B3 race · B4 double-tap · B5 ownership u=1 id · B6 list 0/1/many · B7 token), point analysis, 10 BVA TCs `TC-FR20-BVA-01…10` — immediate (≤1 s) vs delayed (5 s) re-cancel, fresh-shipping recency, parallel cancel‖ship ★, Δt≤300 ms double-tap ★, id±1 sweep, garbage ids (never-500), exactly-1 vs 15-order lists, bad-token 401. Duplicates of DOM avoided (0-list stays DOM-12, guest stays DOM-08, confirmed-cancel stays DOM-02).
- Summaries: FR-20 Designed 18 → 28; Total 165 / Executed 137 / Passed 60 / Failed 77 / Bugs 30.
- No mobile/backend source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Race/double-tap TCs demand ms-precision timing; ownership sweep covers adjacent + garbage ids; list edges avoid DOM-held cells. Review: all timing/ownership/list options approved, hybrid no-phone execution approved; design TCs unchanged.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2–4.3 (boundary + state-transition techniques compose: edges of validity, timing edges, and input-shape edges each get their own TCs) and the skill's step-unit rule: re-cancel is split into immediate vs delayed because the implementation could guard on state alone (both reject) or on state+idempotency-token (immediate double-tap slips through) — only the timing pair distinguishes them.

(5) Student Fix

- Student to review at next gate: confirm race/double-tap timing demands, ownership sweep, list edges, then approve opening mobile/order source for Step 3 (Gap Analysis).
- Review done 16/10 (all approved, hybrid plan confirmed); Artifact 14 verdict confirmed VALID.

Artifact 15: FR-20 white-box inspection + AI Gap Analysis (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~15:40 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3A — open order/mobile source (cancel + history endpoints, ownership checks, real transition matrix; OrderHistory/Detail screens, Cancel handler debounce/dialog/feedback); diff allowed-cancel states (`confirmed` ? `shipping` ? `delivered` ?), integrity on cancel (stock? coupon `used_count` ? transaction?), IDOR, mobile double-tap/dialog; blind spots → `TC-FR20-GAP-*` ; write §5.4; finalize Artifact 14; log Artifact 15; commit `docs(FR-20): complete white-box inspection and AI gap analysis` .

(2) AI Output

- `Main_Report.md` §5.4: implementation table (13 rows with file:line refs), 8 misses M1–M8 with root causes, 10 supplementary TCs `TC-FR20-GAP-01…10` .
- Headline findings: shipping-cancel ALLOWED (comment admits should-be pending/confirmed — server.js:328-331); zero cancel side effects (no stock system, no coupon refund, no txn); mobile fires PUT with no confirm/debounce/loading (App.js:312-329); button hidden unless pending/confirmed (UI guard correct); Vietnamese labels ✓ but no colors, no empty-state illustration, `"Thoát"` label, hardcoded LAN API base; currency bytes verified clean UTF-8 ₫ (hypothesis retracted by evidence).
- No execution performed yet; no Actual/Status filled.

(3) Verdict

VALID — reviewed and confirmed by execution, 16/10/2026. All 8 misses M1–M8 adjudicated at runtime or by exact code refs (shipping-cancel, no refund, no dialog, no colors, text-only empty state, no loading, card/label gaps, hardcoded base); race/double-tap behaved safely by server state (400s), bounding M4 as standard-half-met rather than defect; 404-masking recorded as acceptable reject.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.3 and the Gap-Analysis rubric: mobile and backend are analyzed as two layers of one feature (API-truth vs UI-truth table in §5.4.1 lets execution mark layer-divergent verdicts like DOM-04/UI-vs-API); the retracted mojibake hypothesis is kept in the audit trail as proof that white-box claims require byte-level grounding.

(5) Student Fix

- Student to review at next gate: confirm GAP TC wording, then approve Chặng 3B (execution: API script + Expo-web/viewport screenshots).
- Execution confirms Artifact 15 in full (16/10/2026): 8 bugs FR20-BUG-01…08 filed in `Bug_Reports.md` + `Main_Report.md` §5.5. Runtime extras beyond white-box reading: (a) race ends last-writer-wins with both 200s (TOCTOU amplifier, no torn state); (b) bad-token → 403 not 401 (noted, not filed); (c) Expo-web declined for documented SUT reasons — UI verdicts rest on code refs.

Artifact 16: FR-20 execution + screenshots + bug reports (G9.3 Analyse)

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~16:00 16/10/2026 (UTC+07)
- **Full Prompt (user):** Chặng 3B — execute all FR-20 TCs on live SUT: `scripts/exec_fr20.js` (transitions, IDOR, token, concurrency/double-tap); mobile web/viewport UI screenshots (`FR20_UI_*.png`); fill Actual/Status + summaries; per-bug screenshots; bugs into `Bug_Reports.md` + §5.5; log Artifact 16; commit `test(FR-20): execute test cases, capture screenshots, and document bug reports`.

(2) AI Output

- `scripts/exec_fr20.js` (25 steps; mobA/B/C/D users; admin-driven state setup; parallel race + double-tap; id±1/garbage sweeps; 1- and 15-order histories; bad-token probe; test orders deleted, F9 = 0 left) → `scripts/exec_fr20_results.json`; `scripts/fr20_evidence.html` → `screenshots/FR20_API_EVIDENCE.png` (measured evidence + code-ref panel).
- No `FR20_UI_*.png` : Expo-web boot requires react-native-web/react-dom (absent from SUT deps — installing = SUT modification, declined) and the app hardcodes unreachable 192.168.10.13; mobile UI verdicts grounded in exact App.js line refs instead (disclosed in §5.5 + evidence footer).
- `Main_Report.md` §5.2.3/§5.3.3/§5.4.3: all 38 TCs filled — **24 PASSED, 14 FAILED**; §5.5: 8 bugs FR20-BUG-01…08; summaries (Main + README): Designed 175 / Executed 175 / Passed 84 / Failed 91 / Bugs 38.
- `Bug_Reports.md` : 8 issues (env, pre-conditions, repro, expected/actual, severity, evidence, TC links). GitHub Issue links left TODO (group repo unknown — student files with PNG).

(3) Verdict

VALID (provisional — pending student review before finalization). API statuses derive from measured transcript values (quoted verbatim); every FAILED maps to a filed bug; UI verdicts cite exact code lines (no UI claim without a ref); the missing Expo screenshots are disclosed with reasons, not silently skipped; cleanup verified (0 test orders left).

(4) Reasoning

Per ISTQB CTFL v4.0 §5 and HW02 bug-reporting rules: lifecycle verdicts rest on DB-observed end-states (not response codes alone — the race TC proves why); the evidence split (live transcript vs code refs) is labeled per TC and per bug, so the report never presents a code reading as a runtime measurement.

(5) Student Fix

- Student to review at finalization gate: confirm severity set (esp. BUG-01/02 High calls, UI Lows), file the 8 issues on the group repo and paste links, shoot Expo-Go footage on a real device for the demo video, then approve Giai đoạn Hoàn thiện (Critique, git log, zip).
- Finalization done 16/10: BUG-01 → Critical applied; Critique (243 words) + Accuracy Summary + Disclosure finalized above; GitHub links + demo video remain student TODOs.
- Execution confirms Artifact 11 in full (16/10/2026): 10 bugs FR16-BUG-01…10 filed in `Bug_Reports.md` + `Main_Report.md` §4.5. Runtime extras beyond white-box reading: (a) NULL-price acceptance (missing key → NULL, not just unvalidated values); (b) whitespace-name leftover id 17 (cleanup gap closed manually, script hardened with `TRIM(name) = ''`); (c) 100-row batch in 1684ms (no perf issue — scale TC passes).

Artifact 6: FR-09 BVA black-box (G9.2 Apply) — via domain-bva-tester Step 2

(1) Prompt + Tool

- **Tool:** Muse Spark 1.3 Free (via OpenCode agent, Build Mode)
- **Timestamp:** ~12:20 16/10/2026 (UTC+07)
- **Full Prompt (user):** BVA for FR-09 (still black-box): money edges with u = 1 VND for SAVE10/BIGBUY/VIP100 mins, usage edges + increment-timing probe, expiry edges u = 1s, zero-floor edges; BVA table + `TC-FR09-BVA-*` into §3.3; summaries; finalize Artifact 5; log Artifact 6; commit `feat(FR-09): add Boundary Value Analysis (BVA) points and test cases`. (User-supplied mins SAVE10 100k / VIP100 200k — see correction below.)

(2) AI Output

- **Threshold correction:** user's BVA prompt stated SAVE10 min 100 000 and VIP100 min 200 000 — contradicted by SRS v2.0 (300 000 / 300 000) and seed data (`database.js`: SAVE10 300000, VIP100 300000, BIGBUY 500000, EXPIRED 100000). Corrected to spec values throughout; correction banner placed at §3.3 head. (The 100 000 figure matches EXPIRED's min — likely source of the mix-up.)
- `Main_Report.md` §3.2: DOM-15/16 upgraded to normative single-code-per-order expectation.
- `Main_Report.md` §3.3: 4 boundaries (B1 money u=1₫ · B2 usage u=1 · B3 time u=1s · B4 zero floor), point-analysis tables, 20 BVA TCs `TC-FR09-BVA-01…20` — per-code min−/min/min+ with exact expected money (e.g. 300 000 → −30 000 → 270 000), usage 0/1/2 + apply-without-checkout quota probe (BVA-14 ★), expiry T−1/T/T+1 with `<` vs `<=` probe (BVA-16 ★), floor discount−1/=/+1 (BVA-18…20). Fixture-dependent TCs declare setup + restore.
- Summaries: FR-09 Designed 17 → 37; Total 74 / Executed 37 / Passed 13 / Failed 24 / Bugs 10.
- No coupon/checkout source inspected; all Status = Pending Execution.

(3) Verdict

VALID — reviewed and approved by student, 16/10/2026. Every ordered variable carries an explicit step unit; min-points assert exact money math (not just accept/reject), making `>=` vs `>` failures self-evidencing; the two ★ probes are determination probes per the skill; seed-impossible edges use declared fixtures. Review decisions: spec-min banner kept as verification evidence; B3/B4 fixtures approved with restore; BVA-14/16 kept; source opening authorized.

(4) Reasoning

Per ISTQB CTFL v4.0 §4.2 (BVA at edges with the smallest meaningful increment) and the skill's step-unit rule: u = 1 ₫ is the DB/API integer precision, so min−/min pairs (299 999/300 000) decide the `>=` vs `>` operator with no ambiguity; keeping the user's wrong thresholds would have produced false failures at execution — correcting against the SRS before designing edges is exactly the evidence-before-synthesis discipline the course requires.

(5) Student Fix

- Student to review at next gate: confirm spec-min correction (SAVE10/VIP100 = 300k), fixture plan (cheap items, near-term-expiry coupon, FIX200), probe TCs BVA-14/16, then approve opening coupon/checkout source for Step 3 (Gap Analysis).

AI Accuracy & Gap Analysis Summary (all 4 features)

#	Artifact	Verdict
0	HW02 init (structure + skill)	VALID
1–2	FR-02 Domain + BVA (black-box)	VALID
3–4	FR-02 Gap analysis + execution	VALID
5–6	FR-09 Domain + BVA (black-box)	VALID
7–8	FR-09 Gap analysis + execution	VALID

9–10	FR-16 Domain + BVA (black-box)	VALID
11–12	FR-16 Gap analysis + execution	VALID
13–14	FR-20 Domain + BVA (black-box)	VALID
15–16	FR-20 Gap analysis + execution	VALID

Accuracy ratio: 17/17 VALID (100%), 0 INVALID, 0 INCOMPLETE — every artifact passed student review against ISTQB grounding before its gate commit. (Verdicts measure reviewed-and-corrected outputs, not raw first drafts: the Gap-Analysis sections record what raw AI output missed.)

Miss pattern across features (raw-AI blind spots, all caught by process): spec-faithful operators/constants assumed, never verified (`+2`, `180s`, `>`, percent formula); response shapes and client→server trust never probed (password leak, 1₫ order, honor-system quota); GUI requirements skipped on API-framed features; architecture assumed instead of checked (client-side CSV parse, no-transaction import); layer attribution wrong (trim/normalization).

Heuristic — WHEN to use / not use AI in testing: USE for breadth and bookkeeping — partition enumeration, boundary sweeps with explicit step units, table drafting, audit-trail maintenance, cross-TC consistency checks. DO NOT USE unsupervised for verdicts, operator-level claims (`>=` vs `>`), constants/durations, security conclusions, or any statement about code not yet opened — those require file:line citations, millisecond timestamps, or database recounts. Rule of thumb applied here: no test case leaves `Pending Execution` on AI authority; no bug is filed without a measured transcript step or an exact code reference.

AI Critique (243 words, MANDATORY)

> *"Across four features, the AI proved a fast but credulous assistant: fluent at structuring partitions, boundaries, and tables, yet systematically over-trusting. Its signature failure was assuming a sane implementation behind every spec sentence. It never questioned the increment operator (missed the +2 lockout counter), the lockout constant (missed 180 seconds versus 30), comparison operators (missed strict greater-than on coupon minima), or arithmetic itself (missed a percent formula that inflated totals tenfold). Every miss shares one root cause: the model interpolates the most plausible implementation instead of treating code as an independent, possibly hostile witness — which is precisely why the black-box-first, white-box-second discipline of this assignment matters.*

> *Second, the AI stopped at status codes and messages while ignoring response shapes and trust boundaries. It never probed what else a 200 response carries (plaintext passwords), whether the server recomputes client-supplied totals (it stored a 1-dong order), or whether quota enforcement survives a skipped client call (it does not). Third, it framed each feature as API-only, skipping interface requirements stated in the same specification: text-type password fields, wrong headings, missing status colors.*

> *The collaboration principle I take away: use AI for breadth and bookkeeping — partitions, edge enumeration, audit trails — but reserve every verdict for measured evidence, namely file-and-line citations, millisecond timestamps, and database recounts. The spec corrections recorded in this report (coupon minima, confirmed-cancel validity) show the human's real job: the AI drafts, the tester verifies, and only measurements close a test case."*

> *(Word count: 243 — verified by script, within the required 200–300.)*

Mandatory Disclosure

> *"Test cases, boundary tables, and gap analyses for FR-02/FR-09/FR-16/FR-20 were initially drafted with Muse Spark 1.3 Free (via OpenCode agent); I reviewed every artifact against the EShop SRS and ISTQB CTFL v4.0, corrected operators/thresholds/severities (e.g. coupon minima 300k, confirmed→cancel validity, BUG severity set), added edge cases and white-box probes the AI missed, and personally decided all pass/fail verdicts from measured execution (scripts + transcripts + screenshots in this archive). Screenshots of rendered pages are unaltered captures; the API evidence panels render genuine

> measured output (method stated in each file). YouTube demo footage (when recorded) is my own execution with my own narration. The detailed AI Audit Report above logs every interaction. Group GitHub Issue links (filed 16/10/2026): FR-02 #1–10 · FR-09 #11–20 · FR-16 #21–30 · FR-20 #31–38 — https://github.com/tinphan247/-KTPM-2026-HW02-Domain-Testing/issues. I confirm I did not use AI to generate any artifact in a prohibited category."*

> *Student: Phan Trung Tin — 23120372 — Date: 16/10/2026*